

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

SATURDAY, 1 OCTOBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) questions in Section B, marks will only be awarded for the FIRST questions in the order the candidate submits the answers.

No calculator is allowed for this paper.

Answer questions only in the answer booklet provided.

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION A**

Q1. (a) The Basic JavaFX Program questions.

- (i) When a JavaFX application is launched and, the start method is called, what **object** is passed to the **start** method, and what need to be added to the **object**? (2 marks)
- (ii) Write code fragments to create a rectangle of **width** 150 and **height** 100 with the **upper-left corner** at (20, 10) and fill the rectangle with **green** color. (2 marks)
- (iii) State *THREE* (3) life cycle methods of JavaFX applications. (3 marks)

(b) Correct the following JavaFX code. You may omit the **import** statement in your answer. (3 marks)

```
public void start(Stage primaryStage) {

    // Create a polygon and place it in the scene
    Scene scene = new Scene(new Polygon(), 400, 400);

    // Place the scene in the stage
    primaryStage.setScene(scene);

    // Display the stage
    primaryStage.show();
}
```

- (c) Write a simple JavaFX application to display a message in a window as shown in **Figure 1 (a)**. User can use the radio buttons to change the color of the displayed message. The radio button is shown in **Figure 1 (a)**. For example, the message will become green after the *Green* radio button is selected as shown in **Figure 1(b)**.

Complete the program by filling out the missing codes for parts: **(i), (ii), (iii), (iv)** and **(v)**.

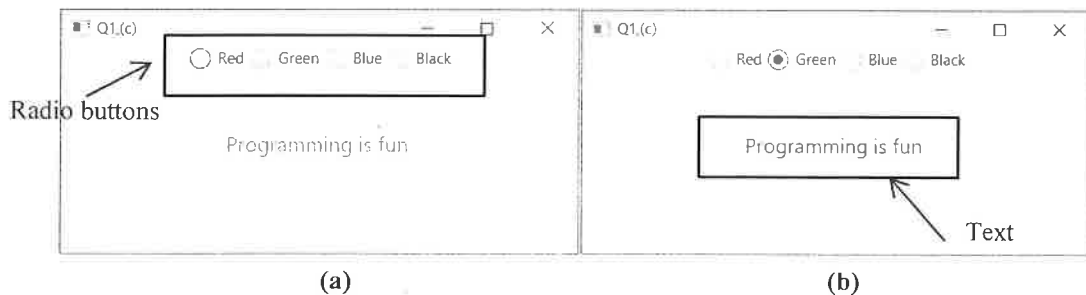


Figure 1

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

```

public class Q1_c extends Application {

    public void start(Stage primaryStage) {

        // refer to (i)

        RadioButton rbRed = new RadioButton("Red");
        RadioButton rbGreen = new RadioButton("Green");
        RadioButton rbBlue = new RadioButton("Blue");
        RadioButton rbBlack = new RadioButton("Black");

        // refer to (ii)

        HBox paneForRadioButtons = new HBox(5);
        paneForRadioButtons.getChildren().addAll(rbRed, rbGreen,
        rbBlue, rbBlack);
        paneForRadioButtons.setAlignment(Pos.CENTER);

        // refer to (iii)

        BorderPane borderPane = new BorderPane();
        borderPane.setTop(paneForRadioButtons);
        borderPane.setCenter(paneForText);

        // refer to (iv)

        // refer to (v)
    }
}

```

- (i) Create a text with specified text (**Programming is fun**) at (160, 80). Fill the text with **gray** color. Set the font type to "*Courier*" and size **20**. You need to use a pane to hold the text. (5 marks)
- (ii) Group the four radio buttons together so that only one button can be selected at a time. (3 marks)
- (iii) Add the event handling code for the four radio buttons to **change the color** for the displayed message as indicated by the name of the radio button using *Lambda expression*. (4 marks)
- (iv) Create a scene of size (500,200) to host the border pane, place the scene into the stage, and show the stage. (2 marks)
- (v) Add `main()` to launch the application. (1 mark)

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

- Q2. (a) The following questions are based on File Input and Output:
(Please include **import** statement in your coding)

A text file "bill.txt" stores item information of the purchase including item ID (String), item category (String), unit price (double), and number of unit (int) as shown below:

DB8546	Appliances	100.00	2
DB5432	Appliances	150.00	1
AT2298	Electronics	300.00	1
DB9912	Appliances	219.50	2
FC3876	Book	78.00	6

Write a Java program to read from the text file and output the TOTAL price (unit price * number of unit) of all the **Appliances** items. For example, your program should output 789.0 (100 x 2 + 150 x 1 + 219.50 x 2) from the records shown above. (11 marks)

- (b) What kind of information may be obtained from a **DatabaseMetaData** object? (2 marks)
- (c) Write a complete java program (JDBC) to:
(Please include **import** statement in your coding.)
- (i) Create an **Employee table** (*id int, sName varchar (20), position varchar(20), dept varchar(20), primary key(id)*) in database *Drive:/DatabaseDir/DatabaseLab.accdb*. (4 marks)
- (ii) Insert values into the table and display the records as shown in **Output** below. (8 marks)

Output:			
Successfully connected to database			
Table creation process successfully!			
122	David	Engineer	Safety
132	John	Clerk	HR
143	James	Operator	QC

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

Q3. (a) What is the key benefit of exception handling in Java? What does the JVM do when an exception occurs? (3 marks)

(b) Analyze the following code and show the output refer to (i), (ii) and (iii):

```
public class Q3_b {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        try {
            System.out.print("Enter an integer: ");
            int number = input.nextInt();
            System.out.println("Result = " + 20/number);
        }
        catch(ArithmeticException e) {
            System.out.println("Arithmetic Exception");
            throw e;
        }
        catch (InputMismatchException e) {
            System.out.println("InputMismatch Exception");
        }
        catch(NumberFormatException e) {
            System.out.println("NumberFormatException");
        }
        finally {
            System.out.println("finally block");
            input.close();
        }
        System.out.println("program ended");
    }
}
```

(i) What is the output when the user enters 5? (3 marks)

(ii) What is the output when the user enters 5.0? (3 marks)

(iii) What is the output when the user enters 0? (3 marks)

(c) State the used of the keyword **throw** and **throws**. (2 marks)

(d) How many ways to create a thread? Briefly explain the steps needed in creating a thread. (6 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q3. (Continued)**

- (e) Identify and correct the error in the following program. (2 marks)

```
1. public class MyTaskClass implements Runnable {  
2.     public static void main(String[] args) {  
3.         new MyTaskClass();  
4.     }  
5.     public MyTaskClass() {  
6.         new Thread(task).run();  
7.     }  
8.     public void run() {  
9.         System.out.println("test");  
10.    }  
11. }
```

- (f) Analyze the following code:

```
public class Count {  
    int count;  
    public void addMethod() {  
        count++;  
    }  
}
```

If two threads try to access the value at the same time, will it cause conflicts?
Explain your answer and suggest if any changes needed. (3 marks)

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION B**

This section consists of TWO (2) questions. Answer **ONLY ONE (1)** question.

- Q4. (a) Describe why `main()` function must be `public`, `static` and `void`. (3 marks)
- (b) Explain the differences between abstract classes and interfaces in terms of data fields, constructors, and methods? (6 marks)
- (c) State the four pillars of object-oriented programming. (4 marks)
- (d) Answer the following sub-questions based on the Java class below:

```

1. public class Q4_d {
2.     public static void main(String[] args) {
3.         int[] arr = {10,20,30,40,50};
4.         MyArray myArr = new MyArray(arr);
5.         arr[0] = 100;
6.         System.out.println(arr[0]);
7.         System.out.println(myArr.getItem(0));
8.     }
9. }
10.
11. class MyArray {
12.     private int[] array;
13.     public MyArray(int[] arr) {
14.         array = arr;
15.     }
16.     public int getItem(int index) {
17.         return array[index];
18.     }
19. }

```

- (i) Explain how encapsulation is implemented in Java. (3 marks)
- (ii) What are the outputs of lines 6 and 7? Justify the obtained results and explain how it violates the concept of encapsulation. (4 marks)
- (iii) Rewrite the constructor of `MyArray` class to prevent such violation from happening. (3 marks)
- (iv) What are the outputs of lines 6 and 7 after rewriting the constructor of `MyArray` class in Q4(c)(iii)? (2 marks)

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

- Q5. (a) A class has static variable and instance variable. Differentiate the two types of variables and describe how they are referenced differently in a class. (4 marks)
- (b) Create a **Circle** class to represent the circle objects, which contains the following requirements. Write your answer in a single Java class.
- (i) Two private data fields named *radius* of double type and *color* of String type, with default value of **2.0** and **"blue"**, respectively. (2 marks)
 - (ii) Two *overloaded* constructors: The no-arg constructor is used to set both radius and color to default value and the second constructor is used to initialize the two data fields (4 marks)
 - (iii) A `getRadius()` method that returns the radius of this circle. A `getArea()` method returns the area of this circle. (4 marks)
 - (iv) Draw the UML class diagram for the Circle class in Q5(b)(i)-(iii). (4 marks)
- (c) Write a test application program that creates two Circle objects by using the two overloaded constructors. Assign default values to first Circle object and assign radius 10.0 and yellow color to the second object. Display the objects and their radius, color and area as shown below.

//Expected output

The first Circle object

The circle has radius of 2.0, color of blue, and area of 12.57

The second Circle object

The circle has radius of 10.0, color of yellow, and area of 314.16

(7 marks)

[Total : 25 marks]